Software Engineering & Project Management (SEPM)

Energy & Utility Management System (EUMS)

Billing, metering and customer service for electricity, water and gas

Project Plan

Software process model: Waterfall · Combined SEPM + DBMS course project

Group number	Group __
Course and section	Software Engineering & Project Management (SEPM), Section __
Supervisor / instructor	Prof. _____
Institute	_____ Institute of Technology
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#	Name	Registration number
1	Sumit P Surendran	20243590
2	Sreethan Reddy	20243550
3	Sushrut Lad	20243551
4	Viswajith	20243559

1. Project Plan

1.1 Original project plan

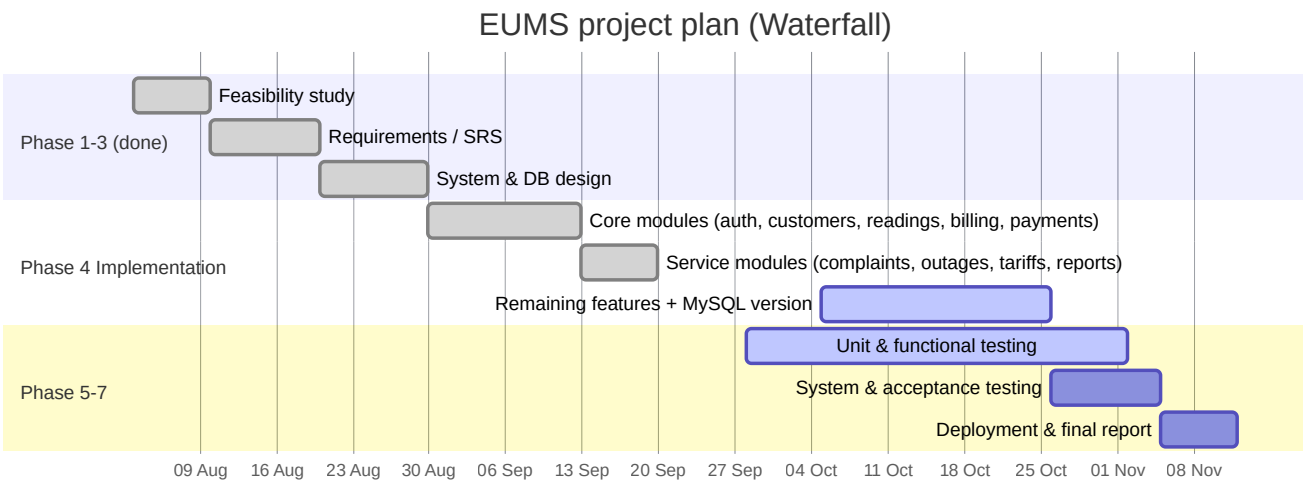


Figure 1: Project schedule (Gantt chart) following the Waterfall phases

Week	Phase	Planned activities	Deliverable
1	Feasibility	Topic study, existing systems, technical/economic/operational/schedule feasibility	Feasibility report
2–3	Requirements	Stakeholders, functional and non-functional requirements, MoSCoW, use cases, user stories	SRS v1.0
4–5	Design	Architecture, ER diagram, normalised schema, triggers, UML, UI mock-up	Design document, schema.sql
5–7	Implementation I	Auth, customers, connections, readings, billing, payments	Working core (end-to-end feature)
8	Implementation II	Complaints, outages, tariffs, dashboards, notifications	Mid-semester build
8–9	Testing I	Unit and functional tests for completed modules	Test report v1
9–11	Implementation III	Reports, meter replacement, late fees, PDF invoice, MySQL port, CSRF	Feature-complete build
12–13	Testing II	System testing, UAT, performance test, bug fixing	Final test report
14	Deployment	Installation guide, final report, demo	Final submission

Table 1: Week-wise plan

1.2 Effort and schedule estimation (Basic COCOMO)

The current build has 2,418 lines of code, so the finished system is estimated at about **3.5 KLOC**. It is a small project built by a small team with familiar requirements, so we use the **organic** mode (a = 2.4, b = 1.05, c = 2.5, d = 0.38):

$$\text{Effort } E = 2.4 \times (3.5)^{1.05} = \mathbf{8.9\ person\text{-}months} \quad \text{Duration } D = 2.5 \times E^{0.38} = \mathbf{5.7\ months} \quad \text{Staff} = E / D \approx \mathbf{1.6\ full\text{-}time\ people}$$

Four students working part-time (about 6 hours a week each, roughly 0.15 of a full-time person) give about 0.6 full-time equivalents over 3.5 months, which is about 2 person-months. That is much less than the COCOMO estimate. The gap is expected: COCOMO was calibrated on industrial projects that include full documentation, formal reviews and maintenance, and it overestimates very small projects that use a high-

level framework. We use it as an upper bound and manage the gap through MoSCoW scoping (Should/Could features can be dropped without breaking the system).